

Supplementary material

Sparse-Concept Calibration of Knowledge Tracing Models for Threshold-Based Educational Decisions

IJIED review copy. Not a classroom trial. ASSISTments locked cells unchanged.

These tables accompany the main 8-page manuscript (Tables S1–S4 and the cluster-aware regression). They are not printed in the main PDF.

Table 1: Four-unique-partition event-level calibration by train-only frequency stratum (same aggregation and $M = 15$ equal-width bins as Table 4). N is the mean test-event count across four unique learner partitions. Flags: Reliable (R) $N \geq 1000$; Limited (L) $100 \leq N < 1000$. Junyi sparse is empty. T-KT is the local Transformer KT baseline. Brier = UNC – RES + REL is the binned decomposition already used in the manuscript; components need not sum exactly to Brier. IRT RES = 0 on these learner-based strata. Junyi T-KT dense N (3,232,614) differs from IRT/DKT (3,226,541) because those models were scored on separately processed test files; ASSISTments and XES3G5M share N across models.

Dataset	Model	Stratum	N	Flag	ECE	Brier	UNC	REL	RES
ASSISTments 2012	IRT	dense	523,971	R	0.0031 $\pm$ 0.0006	0.2103 $\pm$ 0.0003	0.2103 $\pm$ 0.0003	0.0000 $\pm$ 0.0000	0.0000
ASSISTments 2012	IRT	medium	5,963	R	0.1598 $\pm$ 0.0190	0.2742 $\pm$ 0.0075	0.2484 $\pm$ 0.0013	0.0258 $\pm$ 0.0062	0.0000
ASSISTments 2012	IRT	sparse	415	L	0.0777 $\pm$ 0.0555	0.2420 $\pm$ 0.0218	0.2337 $\pm$ 0.0129	0.0084 $\pm$ 0.0090	0.0000
ASSISTments 2012	DKT	dense	523,971	R	0.0602 $\pm$ 0.0022	0.1931 $\pm$ 0.0010	0.2103 $\pm$ 0.0003	0.0053 $\pm$ 0.0005	0.0223 $\pm$ 0.0004
ASSISTments 2012	DKT	medium	5,963	R	0.1621 $\pm$ 0.0256	0.2297 $\pm$ 0.0144	0.2484 $\pm$ 0.0013	0.0301 $\pm$ 0.0086	0.0479 $\pm$ 0.0053
ASSISTments 2012	DKT	sparse	415	L	0.2333 $\pm$ 0.0084	0.2502 $\pm$ 0.0098	0.2337 $\pm$ 0.0129	0.0624 $\pm$ 0.0038	0.0451 $\pm$ 0.0037
ASSISTments 2012	T-KT	dense	523,971	R	0.1136 $\pm$ 0.0066	0.2116 $\pm$ 0.0019	0.2103 $\pm$ 0.0003	0.0201 $\pm$ 0.0015	0.0186 $\pm$ 0.0004
ASSISTments 2012	T-KT	medium	5,963	R	0.1541 $\pm$ 0.0051	0.2272 $\pm$ 0.0086	0.2484 $\pm$ 0.0013	0.0273 $\pm$ 0.0021	0.0478 $\pm$ 0.0053
ASSISTments 2012	T-KT	sparse	415	L	0.2280 $\pm$ 0.0197	0.2525 $\pm$ 0.0116	0.2337 $\pm$ 0.0129	0.0594 $\pm$ 0.0104	0.0392 $\pm$ 0.0088
Junyi Academy	IRT	dense	3,226,541	R	0.0026 $\pm$ 0.0018	0.2078 $\pm$ 0.0007	0.2078 $\pm$ 0.0007	0.0000 $\pm$ 0.0000	0.0000
Junyi Academy	IRT	medium	3,706	R	0.1209 $\pm$ 0.0082	0.2579 $\pm$ 0.0033	0.2432 $\pm$ 0.0013	0.0147 $\pm$ 0.0021	0.0000
Junyi Academy	DKT	dense	3,226,541	R	0.0389 $\pm$ 0.0022	0.1804 $\pm$ 0.0008	0.2078 $\pm$ 0.0007	0.0022 $\pm$ 0.0002	0.0294 $\pm$ 0.0002
Junyi Academy	DKT	medium	3,706	R	0.2513 $\pm$ 0.0217	0.3077 $\pm$ 0.0144	0.2432 $\pm$ 0.0013	0.0741 $\pm$ 0.0123	0.0091 $\pm$ 0.0033
Junyi Academy	T-KT	dense	3,232,614	R	0.0792 $\pm$ 0.0051	0.1884 $\pm$ 0.0020	0.2079 $\pm$ 0.0003	0.0084 $\pm$ 0.0015	0.0279 $\pm$ 0.0008
Junyi Academy	T-KT	medium	3,836	R	0.1073 $\pm$ 0.0156	0.2307 $\pm$ 0.0103	0.2417 $\pm$ 0.0038	0.0149 $\pm$ 0.0046	0.0260 $\pm$ 0.0058
XES3G5M	IRT	dense	1,269,345	R	0.0025 $\pm$ 0.0022	0.1622 $\pm$ 0.0018	0.1622 $\pm$ 0.0018	0.0000 $\pm$ 0.0000	0.0000
XES3G5M	IRT	medium	12,889	R	0.0122 $\pm$ 0.0062	0.1696 $\pm$ 0.0034	0.1695 $\pm$ 0.0032	0.0002 $\pm$ 0.0001	0.0000
XES3G5M	IRT	sparse	1,969	R	0.0424 $\pm$ 0.0097	0.1875 $\pm$ 0.0055	0.1856 $\pm$ 0.0048	0.0019 $\pm$ 0.0008	0.0000
XES3G5M	DKT	dense	1,269,345	R	0.0378 $\pm$ 0.0005	0.1223 $\pm$ 0.0009	0.1622 $\pm$ 0.0018	0.0027 $\pm$ 0.0002	0.0420 $\pm$ 0.0009
XES3G5M	DKT	medium	12,889	R	0.0788 $\pm$ 0.0013	0.1186 $\pm$ 0.0025	0.1695 $\pm$ 0.0032	0.0102 $\pm$ 0.0008	0.0601 $\pm$ 0.0016
XES3G5M	DKT	sparse	1,969	R	0.1139 $\pm$ 0.0054	0.1391 $\pm$ 0.0063	0.1856 $\pm$ 0.0048	0.0180 $\pm$ 0.0011	0.0632 $\pm$ 0.0040
XES3G5M	T-KT	dense	1,269,345	R	0.1176 $\pm$ 0.0014	0.1562 $\pm$ 0.0018	0.1622 $\pm$ 0.0018	0.0184 $\pm$ 0.0005	0.0236 $\pm$ 0.0006
XES3G5M	T-KT	medium	12,889	R	0.1129 $\pm$ 0.0061	0.1366 $\pm$ 0.0051	0.1695 $\pm$ 0.0032	0.0176 $\pm$ 0.0018	0.0493 $\pm$ 0.0010
XES3G5M	T-KT	sparse	1,969	R	0.1254 $\pm$ 0.0047	0.1493 $\pm$ 0.0043	0.1856 $\pm$ 0.0048	0.0216 $\pm$ 0.0023	0.0567 $\pm$ 0.0018

Table 2: Complete within-KC controlled sparsification (seed 42; 30 originally dense KCs per dataset; learner-based fold 0). Δ = reduced–full. Positive Δ ECE means worse calibration after reducing training rows for the same KC. 95% CIs are bootstrap over KCs. T-KT is the local Transformer KT baseline, not published SimpleKT. This table reports every dataset $\times$ model $\times$ {500, 100, 50} cell; none is omitted. Not a causal law for real-world sparsity.

Dataset	Model	Rows	Δ ECE [95% CI]	Δ REL [95% CI]	Share ECE $\uparrow$	n_{KC}
ASSISTments 2012	DKT	500	−0.047 [−0.060, −0.033]	−0.015 [−0.022, −0.010]	10%	30
ASSISTments 2012	DKT	100	−0.024 [−0.046, −0.001]	−0.008 [−0.017, +0.001]	30%	30
ASSISTments 2012	DKT	50	−0.017 [−0.034, +0.002]	−0.008 [−0.016, −0.002]	43%	30
ASSISTments 2012	T-KT	500	−0.026 [−0.047, −0.002]	−0.008 [−0.017, +0.003]	20%	30
ASSISTments 2012	T-KT	100	+0.002 [−0.023, +0.029]	0.000 [−0.013, +0.014]	43%	30
ASSISTments 2012	T-KT	50	+0.002 [−0.021, +0.025]	+0.001 [−0.011, +0.012]	47%	30
Junyi Academy	DKT	500	−0.021 [−0.041, −0.001]	−0.007 [−0.013, −0.002]	30%	30
Junyi Academy	DKT	100	−0.010 [−0.031, +0.011]	−0.003 [−0.009, +0.002]	37%	30
Junyi Academy	DKT	50	−0.004 [−0.022, +0.013]	−0.002 [−0.008, +0.003]	57%	30
Junyi Academy	T-KT	500	+0.101 [+0.081, +0.120]	+0.048 [+0.035, +0.060]	93%	30
Junyi Academy	T-KT	100	+0.132 [+0.109, +0.155]	+0.061 [+0.047, +0.077]	100%	30
Junyi Academy	T-KT	50	+0.135 [+0.110, +0.161]	+0.061 [+0.047, +0.074]	100%	30
XES3G5M	DKT	500	+0.142 [+0.104, +0.189]	+0.065 [+0.039, +0.103]	97%	30
XES3G5M	DKT	100	+0.137 [+0.107, +0.170]	+0.059 [+0.040, +0.081]	100%	30
XES3G5M	DKT	50	+0.161 [+0.127, +0.196]	+0.077 [+0.051, +0.106]	100%	30
XES3G5M	T-KT	500	+0.041 [+0.020, +0.063]	+0.021 [+0.010, +0.032]	83%	30
XES3G5M	T-KT	100	+0.083 [+0.061, +0.111]	+0.044 [+0.031, +0.058]	100%	30
XES3G5M	T-KT	50	+0.110 [+0.071, +0.156]	+0.070 [+0.040, +0.106]	93%	30

Table 3: Locked τ grid on ASSISTments 2012, learner-based fold 0 (seed 42), T-KT only. FAR = $P(y = 0 \mid p \geq \tau)$. Sparse occupancy is Limited ($N = 444$). Display threshold in the main text is $\tau = 0.7$ (locked FAR 0.196 / 0.268). The grid is not a search over sparse error.

τ	Dense FAR	Sparse FAR	Δ FAR	Sparse N_{advance}	Sparse flag
0.5	0.234	0.287	0.053	275	L
0.6	0.215	0.275	0.060	244	L
0.7	0.196	0.268	0.072	235	L
0.8	0.177	0.261	0.084	218	L

Table 4: Simulated practice-gate policies on frozen T-KT probabilities, ASSISTments 2012 learner-based fold 0 (seed 42). Same p as Table 5 / Table S3; CSV model frozen `simplekt` is the local T-KT baseline, not published SimpleKT. Policy A: global $\tau = 0.7$. Policy B: advance only if occupancy is Reliable (dense+medium). Policy C: $\tau = 0.8$ on the sparse stratum and $\tau = 0.7$ elsewhere. Population FAR/Miss use all test events ($N = 534,150$). Sparse FAR is undefined when $N_{\text{advance}} = 0$. Not a classroom trial.

Policy	Pop. N_{advance}	Pop. FAR	Pop. Miss	Sparse N_{advance}	Sparse FAR	Sparse Miss
A global $\tau = 0.7$	287,164	0.197	0.350	235	0.268	0.320
B Reliable-only	286,921	0.197	0.350	0	—	0
C sparse $\tau = 0.8$	287,147	0.197	0.350	218	0.261	0.289

Table 5: Cluster-aware T-KT ECE regression (Section IV.D). ASSISTments and Junyi rows are KC-fold observations at fold 0 with cluster-robust SE at `kc_id`. XES3G5M rows are the A2B-masked weighted fit ($n = 829$ unique KCs; padding excluded). Predictors are standardized within dataset. Not a causal frequency effect.

Dataset	Covariate	n_{obs}	Unique KCs	$\hat{\beta}$	SE _{cl}	95% CI	p
<i>Primary: weighted WLS, cluster-robust SE</i>							
ASSISTments 2012	$\log(1 + f_{\text{train}})$	478	261	-0.068	0.009	[-0.086, -0.050]	< 0.001
	Difficulty proxy	478	261	+0.026	0.005	[+0.016, +0.036]	< 0.001
	Item support	478	261	-0.006	0.003	[-0.011, -0.000]	0.046
	Learner exposure	478	261	+0.016	0.006	[+0.004, +0.027]	0.007
	Curriculum position	478	261	-0.005	0.004	[-0.012, +0.003]	0.217
Junyi Academy	$\log(1 + f_{\text{train}})$	2,645	1,326	-0.014	0.002	[-0.018, -0.010]	< 0.001
	Difficulty proxy	2,645	1,326	+0.003	0.001	[+0.001, +0.005]	0.008
	Item support	2,645	1,326	-0.001	0.001	[-0.002, +0.001]	0.280
	Learner exposure	2,645	1,326	+0.001	0.001	[-0.002, +0.003]	0.573
	Curriculum position	2,645	1,326	-0.004	0.001	[-0.005, -0.002]	< 0.001
XES3G5M	$\log(1 + f_{\text{train}})$	829	829	-0.028	0.007	[-0.042, -0.014]	< 0.001
	Difficulty proxy	829	829	+0.080	0.004	[+0.072, +0.088]	< 0.001
	Item support	829	829	-0.007	0.001	[-0.010, -0.005]	< 0.001
	Learner exposure	829	829	+0.013	0.004	[+0.005, +0.021]	0.001
	Curriculum position	829	829	-0.014	0.002	[-0.019, -0.009]	< 0.001
<i>Sensitivity: unweighted OLS, same cluster-robust SE</i>							
ASSISTments 2012	$\log(1 + f_{\text{train}})$	478	261	-0.060	0.014	[-0.086, -0.033]	< 0.001
	Difficulty proxy	478	261	+0.029	0.010	[+0.009, +0.049]	0.004
	Item support	478	261	-0.012	0.003	[-0.019, -0.005]	< 0.001
	Learner exposure	478	261	+0.021	0.007	[+0.007, +0.036]	0.005
	Curriculum position	478	261	-0.009	0.007	[-0.022, +0.004]	0.170
Junyi Academy	$\log(1 + f_{\text{train}})$	2,645	1,326	-0.022	0.002	[-0.026, -0.018]	< 0.001
	Difficulty proxy	2,645	1,326	+0.005	0.001	[+0.004, +0.007]	< 0.001
	Item support	2,645	1,326	-0.000	0.001	[-0.002, +0.001]	0.728
	Learner exposure	2,645	1,326	+0.009	0.002	[+0.004, +0.013]	< 0.001
	Curriculum position	2,645	1,326	-0.001	0.001	[-0.003, +0.000]	0.106
XES3G5M	$\log(1 + f_{\text{train}})$	829	829	-0.008	0.012	[-0.031, +0.015]	0.488
	Difficulty proxy	829	829	+0.070	0.006	[+0.057, +0.083]	< 0.001
	Item support	829	829	-0.009	0.002	[-0.014, -0.004]	< 0.001
	Learner exposure	829	829	+0.007	0.008	[-0.009, +0.023]	0.396
	Curriculum position	829	829	-0.017	0.004	[-0.025, -0.009]	< 0.001